

Ben-Gurion University of the Negev

Faculty of Engineering Sciences

[DEPARTMENT DETAILS REQUIRED]

A DAG-Guided Framework for Proxy-State Effect Estimation in Irregular ICU Time Series

Thesis submitted in partial fulfillment of the requirements for the Master
degree

[AUTHOR DETAILS REQUIRED]

[SUPERVISOR DETAILS REQUIRED]

[ADMINISTRATIVE DETAILS REQUIRED]

Approval Page

A DAG-Guided Framework for Proxy-State Effect Estimation in Irregular ICU Time Series

[AUTHOR DETAILS REQUIRED]

[SUPERVISOR DETAILS REQUIRED]

[DEPARTMENT DETAILS REQUIRED]

[APPROVAL TEXT REQUIRED]

[ADMINISTRATIVE DETAILS REQUIRED]

Secondary-Language Abstract

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Primary-Language Abstract

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Keywords

Primary-Language Keywords

[STAGE 4 DRAFT REQUIRED]

Secondary-Language Keywords

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Acknowledgements

[STAGE 4 DRAFT REQUIRED]

[ADMINISTRATIVE DETAILS REQUIRED]

Contents

Secondary-Language Abstract	i
Primary-Language Abstract	ii
Keywords	iii
Acknowledgements	iv
Abbreviations	viii
Notation and Symbols	ix
1 Introduction	1
1.1 Motivation: Irregular ICU Time-Series Analysis	1
1.2 Thesis Gap and Objective	1
1.3 Research Questions and Contributions	1
1.4 Thesis Organization	1
2 Background and Related Work	2
2.1 ICU EHR Datasets and Irregular Sampling	2
2.2 Irregular Time-Series Representation Models	2
2.3 Proxy Phenotyping and Weak Supervision	2
2.4 Causal Diagrams, Identification, HTE, Overlap, and Sensitivity	2
3 Problem Definition and Study Design	3
3.1 Units, Time Horizons, and Data Objects	3
3.2 Prediction Task	4
3.3 Causal Question, Exposures, Outcome, Estimands, Assumptions	4
4 Data and Preprocessing	6
4.1 PhysioNet 2012 Pipeline	6
4.2 MIMIC-III Pipeline	6
4.3 STraTS Split-Aware Artifacts Versus Causal Artifacts	7

5	Clinical Proxy-State Construction	9
5.1	Rationale and Terminology	9
5.2	PhysioNet Proxy-State Rules	9
5.3	MIMIC Proxy-State Rules and Validation Artifacts	9
5.4	Predicted and Majority-Vote Proxy States	9
6	Predictive Modeling of Proxy States	10
6.1	Self-Supervised STraTS Pretraining	10
6.2	Supervised Multi-Label Models	10
6.3	Prediction Export and Normalization	10
7	Causal Graph and Effect-Estimation Methodology	11
7.1	Dataset-Specific DAGs	11
7.2	Adjustment-Set Logic	11
7.3	Matching and Heterogeneous-Effect Estimators	11
8	Robustness, Sensitivity, and Validation Design	12
8.1	Overlap and Support Diagnostics	12
8.2	Sensitivity and Robustness Values	12
8.3	Permutation Checks and Reproducibility Validation	12
9	Experimental Design	13
9.1	Prediction Experiment Matrix	13
9.2	Causal Experiment Matrix	13
10	Results	14
10.1	Data and Cohort Summary	14
10.2	Proxy Prevalence and Co-Occurrence	14
10.3	Predictive Performance	14
10.4	Learning Curves	14
10.5	Mortality Prediction From Proxy States	14
10.6	Matching Results	15
10.7	CATE Estimates	15
10.8	Heterogeneity Diagnostics	15
10.9	Overlap and Support	15
10.10	Sensitivity	15
10.11	Permutation	15
10.12	Cross-Dataset Comparison	15
11	Discussion	16
11.1	Answer Research Questions	16
11.2	Limitations and Threats to Validity	16

12 Conclusions and Future Work	17
12.1 Contribution Summary and Future Work	17
A Complete Proxy-State Definitions	18
B DAG Node and Edge Inventory	19
C Configuration Fields and Resolved Run Settings	20
D Model Hyperparameters and Training Commands	21
E Additional Figures and Tables	22
F Reproducibility and Environment Notes	23
G Legacy Code and Excluded Artifacts	24
Bibliography	25

Abbreviations

[STAGE 4 DRAFT REQUIRED]

Notation and Symbols

[STAGE 4 DRAFT REQUIRED]

List of Figures

List of Tables

4.1	Processed artifact contracts used by the causal and STraTS repositories.	8
-----	--	---

Chapter 1

Introduction

1.1 Motivation: Irregular ICU Time-Series Analysis

[STAGE 4 DRAFT REQUIRED]

1.2 Thesis Gap and Objective

[STAGE 4 DRAFT REQUIRED]

1.3 Research Questions and Contributions

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

1.4 Thesis Organization

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Chapter 2

Background and Related Work

2.1 ICU EHR Datasets and Irregular Sampling

[STAGE 4 DRAFT REQUIRED]

2.2 Irregular Time-Series Representation Models

[STAGE 4 DRAFT REQUIRED]

2.3 Proxy Phenotyping and Weak Supervision

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

2.4 Causal Diagrams, Identification, HTE, Overlap, and Sensitivity

[STAGE 4 DRAFT REQUIRED]

[CITATION REQUIRED]

Chapter 3

Problem Definition and Study Design

3.1 Units, Time Horizons, and Data Objects

This thesis studies retrospective intensive-care records represented as irregular clinical time series. The operational study unit is a time-series record or ICU stay, depending on the source dataset and preprocessing path. PhysioNet 2012 source files use record identifiers, whereas MIMIC-III source tables use ICU-stay and admission identifiers. The implemented pipelines normalize these source-specific identifiers to the canonical join key `ts_id`. This identifier should be read as the analysis-record identifier used by the software; it is not assumed to be a globally unique person identifier across repeated admissions or stays.

For a study unit i , the observed longitudinal data are represented as a finite collection of measurement events

$$\mathcal{O}_i = \{(i, t_{ij}, v_{ij}, x_{ij}) : j = 1, \dots, n_i\},$$

where i is the normalized `ts_id`, t_{ij} is elapsed time in minutes from record start or ICU admission, $v_{ij} \in \mathcal{V}$ is the observed variable name, and x_{ij} is the numeric value recorded for that variable. In the causal repository this long-format object is stored in `ts` with the required columns `ts_id`, `minute`, `variable`, and `value`. Static or baseline quantities such as age, sex or gender coding, height, weight, and ICU-type indicators are represented within the same event schema as variables observed at or near the beginning of the record when the preprocessing implementation provides them.

The event representation is intentionally irregular. Observations occur at nonuniform times, different variables are measured with different frequencies, and most variables are absent at most possible times. A missing measurement is therefore not interpreted as a normal value. It may reflect a clinical decision not to measure, an unrecorded observation, a source-table limitation, or a preprocessing exclusion. The thesis treats informative measurement and missingness processes as important threats to interpretation, not as causal findings established by the repository.

The primary patient- or stay-level outcome used by the downstream causal repository is `in_hospital_mortality`. In the causal processed artifact, this outcome is stored in `oc`, the outcome table keyed by `ts_id`. The third object in that processed artifact, `ts_ids`, is the sorted collection of normalized identifiers represented in `ts`. Later predictive processing uses a

distinct split-aware artifact and must not be conflated with this causal contract.

3.2 Prediction Task

The prediction task is a multi-label proxy-state prediction problem over irregular ICU time series. For each study unit i and proxy state k , let $L_{ik}^{rule} \in \{0, 1\}$ denote a rule-derived proxy label. These labels are clinically inspired analytical constructs generated from observed data and deterministic source-code rules. They are weak clinical labels or proxy phenotypes, not chart-adjudicated diagnoses, manually verified clinical states, or validated phenotypes unless separate validation evidence is supplied in a later chapter.

The supervised predictive workflow estimates a vector of proxy-state labels from the observed time-series representation. Its target matrix is loaded from a latent-label CSV keyed by `ts_id`; the supervised proxy-label targets are not taken from the mortality column in `oc`. For model m , the exported prediction object may contain probabilities $\hat{p}_{ik}^{(m)}$ and corresponding binary predicted labels $\hat{L}_{ik}^{(m)}$. These outputs are model estimates of rule-derived proxy labels rather than direct measurements of patient physiology.

At the level of study design, five analytical tasks are distinguished. First, rule-based proxy-state construction derives binary proxy labels from observed clinical measurements. Second, multi-label proxy-state prediction estimates those rule-derived labels from irregular time-series inputs. Third, proxy-label aggregation may combine binary outputs from several voter sources into a single proxy-state table. Fourth, mortality prediction or association analysis evaluates whether proxy-state representations contain prognostic information about `in_hospital_mortality`. Fifth, observational causal-effect estimation treats selected proxy-state indicators as modeled exposures and estimates adjusted contrasts under explicit assumptions. Detailed proxy rules, model architectures, estimator algorithms, and numerical results are intentionally deferred to later chapters.

3.3 Causal Question, Exposures, Outcome, Estimands, Assumptions

The downstream causal-analysis setting is retrospective and observational. No intervention is assigned by this study. The repository uses software parameters named `treatment` for selected binary `LAT_*` columns, but many such columns represent illness-state or organ-dysfunction proxy states rather than well-defined administered therapies. In the thesis narrative these variables are therefore described as proxy-state exposures or modeled treatment variables unless a later section explicitly defines a manipulable intervention.

Let $A_i \in \{0, 1\}$ denote a selected proxy-state exposure, Y_i denote the binary outcome `in_hospital_mortality`, W_i denote observed adjustment variables selected from a project-authored DAG and available columns, and X_i denote effect-modifier features used to describe heterogeneity. Potential-outcome notation such as $Y_i(a)$ is useful as conceptual language, but

causal interpretation requires more than the presence of a graph or an estimator. It depends on assumptions about the causal graph, consistency, exchangeability conditional on observed covariates, positivity or overlap, outcome and exposure measurement, and unmeasured confounding [1, 2, 3].

Accordingly, later estimates should be interpreted as adjusted effect estimates under project assumptions, not as unconditional clinical effects. The existence of a DAG, matching script, or CATE estimator does not prove identification. Proxy states may be useful for structured analysis while still carrying label noise, temporal ambiguity, and intervention-definition limitations.

[SUPERVISOR DECISION REQUIRED: final estimand wording for proxy-state exposures and aggregated CATE summaries]

The approved main research question is how irregular ICU time-series data can be converted into clinically interpretable proxy-state representations and used, under explicit causal assumptions, to support DAG-guided adjusted effect estimation for in-hospital mortality. Within the scope of this chapter and the next, the relevant subquestions are whether rule-derived clinical proxy states can be predicted from irregular ICU time series, which data contracts are needed to connect PhysioNet 2012, MIMIC-III, STraTS, and the causal-analysis pipeline, whether proxy-state representations contain mortality-related information, how adjusted estimates vary across proxy-state exposures or patient characteristics, and how robust those estimates are to modeling and confounding assumptions. These are study questions and design commitments; their answers are reserved for the results and discussion chapters.

Chapter 4

Data and Preprocessing

4.1 PhysioNet 2012 Pipeline

The PhysioNet/Computing in Cardiology Challenge 2012 dataset provides ICU time-series records and associated in-hospital mortality outcomes for a mortality-prediction challenge setting [4]. In this thesis it functions as one of the two ICU time-series sources used to exercise the proxy-state prediction and downstream causal-analysis pipeline. The local repository does not track the raw challenge files or a verified processed-data pickle, so this section describes the implemented preprocessing contract rather than asserting final cohort counts.

[RESULT REQUIRED: final number of included PhysioNet records after preprocessing]

The causal-repository PhysioNet preprocessing implementation traverses the raw `set-a`, `set-b`, and `set-c` folders and reads one patient-record file at a time. It removes rows with missing parameter names, skips records with at most five retained measurement rows, and removes observations with negative values, which the source code treats as missingness indicators. It converts source times from HH:MM strings into elapsed minutes, renames `RecordID` to `ts_id`, `Parameter` to `variable`, and `Value` to `value`, and reads outcome files by mapping `In-hospital_death` to `in_hospital_mortality`. After concatenating the source sets, it drops duplicate events and converts the categorical `ICUType` measurement into one-hot-style variables `ICUType_1` through `ICUType_4`.

The resulting causal processed artifact is serialized as three objects: `ts`, `oc`, and `ts_ids`. The `ts` object is a long event table with `ts_id`, `minute`, `variable`, and `value`. The `oc` object contains `ts_id`, `length_of_stay`, `in_hospital_mortality`, and a source subset indicator. The `ts_ids` object is derived from identifiers observed in `ts`. This artifact is the expected input to rule-based tagging, mortality-from-proxy analysis, matching, and CATE estimation in the causal repository.

4.2 MIMIC-III Pipeline

MIMIC-III is a critical-care electronic health-record database used here as the second ICU data source [5]. The implemented workflow follows a clinical time-series benchmark style in that it

transforms raw ICU events into stay-level time-series examples suitable for prediction, while adapting the output contract for this thesis pipeline [6]. The repository does not establish a tracked raw-data snapshot, extraction date, or final processed artifact hash.

[RESULT REQUIRED: final number of included MIMIC-III ICU stays after preprocessing]

The active causal-repository MIMIC preprocessing implementation reads broad categories of MIMIC-III source tables: ICU-stay timing from `ICUSTAYS`, demographics from `PATIENTS`, bedside measurements from `CHARTEVENTS`, laboratory measurements from `LABEVENTS`, fluid outputs from `OUTPUTEVENTS`, administered inputs from `INPUTEVENTS_CV` and `INPUTEVENTS_MV`, and mortality/timing information from `ADMISSIONS`. Large event tables are processed in chunks and written to temporary shards before feature rows are collected. The implementation filters to adult ICU stays, removes chart events marked with an error flag, requires valid chart times and numeric or textual values, applies item-ID mappings and range filters for selected variables, converts selected units, and assigns events without direct ICU-stay identifiers to an ICU interval when possible.

For the causal contract, event times are converted to elapsed minutes relative to ICU admission, events are restricted to the early ICU window used by the implementation, age and gender are added as static rows at minute zero, and duplicate (`ts_id`, `minute`, `variable`) events are collapsed by averaging. The helper contract canonicalizes stay identifiers, normalizing numeric-looking identifiers such as 12345.0 to 12345. It defines the canonical `ts` columns as `ts_id`, `minute`, `variable`, and `value`, defines the canonical `oc` columns as `ts_id`, `length_of_stay`, `in_hospital_mortality`, and `subset`, and intentionally excludes internal MIMIC identifiers such as `HADM_ID` and `SUBJECT_ID` from the exported outcome table.

Inspection found a source-level issue that should be resolved before relying on local MIMIC preprocessing execution: the active script reads `ADMISSIONS.csv` with `DEATHTIME` for early-death filtering, while the canonical outcome helper requires `HOSPITAL_EXPIRE_FLAG` to construct `in_hospital_mortality`. This does not change the documented target contract, but it is a reproducibility risk recorded in the Stage 4.1 evidence report and deferred-fix register.

4.3 STraTS Split-Aware Artifacts Versus Causal Artifacts

The predictive STraTS repository uses a different processed-data contract from the causal repository. Its loader expects a five-object pickle consisting of `events`, `oc`, `train_ids`, `val_ids`, and `test_ids`, where the final three objects explicitly encode train, validation, and test identifiers. The loader accepts stay identifiers named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and validates the split arrays after the same normalization. This split-aware contract is therefore not interchangeable with the causal repository’s unsplit [`ts`, `oc`, `ts_ids`] artifact.

In supervised STraTS runs, the latent-label CSV is joined to the selected split identifiers after ID normalization. The loader filters labels to the supervised IDs, raises an error when

labels are missing for any supervised `ts_id`, raises an error for duplicate normalized labels, and uses all non-`ts_id` columns in the latent CSV as proxy-label targets. The mortality column in `oc` is not used as the supervised proxy-label target. Prediction export writes `ts_id`, one probability column per proxy state, and one thresholded binary column per proxy state; the wrapper scripts inspected for this stage export the `all` split for their prediction CSVs, although final archive-copy provenance remains incomplete.

The predictive loader also implements split-aware leakage controls. It keeps only variables observed in the training split, derives normalization statistics from the training split, computes static-feature normalization from training rows, and validates label alignment before training. For MIMIC-III supervised prediction it restricts events to the first 24 hours. The pretraining path removes test data, uses training-derived variable statistics, uses a 48-hour maximum window for PhysioNet 2012, and permits MIMIC-III observations up to five days while sampling 24-hour input windows. These controls reduce some sources of leakage, but they do not guarantee absence of leakage, nor do they resolve missing raw-data and split-provenance limitations.

The parent router can bridge contracts by reading a causal `[ts, oc, ts_ids]` artifact and constructing a STraTS-style `[ts, oc, train_ids, val_ids, test_ids]` payload through a seeded identifier shuffle. That bridge is an orchestration convenience and does not prove how every archived final STraTS artifact was generated.

Table 4.1: Processed artifact contracts used by the causal and STraTS repositories.

Property	Causal repository	STraTS repository
Processed artifact	<code>[ts, oc, ts_ids]</code>	<code>[events, oc, train_ids, val_ids, test_ids]</code>
Split representation	Unsplit ID collection	Explicit train/validation/test IDs
Main use	Tagging and causal analysis	Predictive training and export
Identifier convention	Canonical <code>ts_id</code>	Loader-normalized <code>ts_id</code>
Proxy-label source	Separate latent CSV down-stream	Separate latent CSV joined to split IDs

[FIGURE REQUIRED: audited dataflow diagram linking raw data, causal artifacts, STraTS artifacts, proxy labels, and causal analysis]

Several provenance limitations remain. Raw PhysioNet and MIMIC-III data are not tracked in Git. The processed pickles used by final runs are external or absent from the audited repository. Some run records contain machine-specific absolute paths, and final cohort counts require verified processed artifacts or manifests. Clean-checkout reproduction therefore requires authorized data access, artifact hashes, and a documented split-generation record.

[VALIDATION REQUIRED: checksums for processed PhysioNet and MIMIC-III dataset artifacts]

[VALIDATION REQUIRED: provenance of train/validation/test split generation for final STraTS artifacts]

Chapter 5

Clinical Proxy-State Construction

5.1 Rationale and Terminology

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

5.2 PhysioNet Proxy-State Rules

[STAGE 4 DRAFT REQUIRED]

[TABLE REQUIRED]

5.3 MIMIC Proxy-State Rules and Validation Artifacts

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

[TABLE REQUIRED]

5.4 Predicted and Majority-Vote Proxy States

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Chapter 6

Predictive Modeling of Proxy States

6.1 Self-Supervised STraTS Pretraining

[STAGE 4 DRAFT REQUIRED]

6.2 Supervised Multi-Label Models

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

6.3 Prediction Export and Normalization

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Chapter 7

Causal Graph and Effect-Estimation Methodology

7.1 Dataset-Specific DAGs

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

7.2 Adjustment-Set Logic

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

7.3 Matching and Heterogeneous-Effect Estimators

[STAGE 4 DRAFT REQUIRED]

[CITATION REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Chapter 8

Robustness, Sensitivity, and Validation Design

8.1 Overlap and Support Diagnostics

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

[VALIDATION REQUIRED]

8.2 Sensitivity and Robustness Values

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

8.3 Permutation Checks and Reproducibility Validation

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Chapter 9

Experimental Design

9.1 Prediction Experiment Matrix

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

[VALIDATION REQUIRED]

9.2 Causal Experiment Matrix

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

[VALIDATION REQUIRED]

Chapter 10

Results

10.1 Data and Cohort Summary

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.2 Proxy Prevalence and Co-Occurrence

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.3 Predictive Performance

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.4 Learning Curves

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.5 Mortality Prediction From Proxy States

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.6 Matching Results

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.7 CATE Estimates

[RESULT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

[VALIDATION REQUIRED]

10.8 Heterogeneity Diagnostics

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.9 Overlap and Support

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.10 Sensitivity

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.11 Permutation

[RESULT REQUIRED]

[VALIDATION REQUIRED]

10.12 Cross-Dataset Comparison

[RESULT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

[VALIDATION REQUIRED]

Chapter 11

Discussion

11.1 Answer Research Questions

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

[VALIDATION REQUIRED]

11.2 Limitations and Threats to Validity

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

[VALIDATION REQUIRED]

Chapter 12

Conclusions and Future Work

12.1 Contribution Summary and Future Work

[STAGE 4 DRAFT REQUIRED]

[RESULT REQUIRED]

Appendix A

Complete Proxy-State Definitions

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Appendix B

DAG Node and Edge Inventory

[STAGE 4 DRAFT REQUIRED]

[SUPERVISOR DECISION REQUIRED]

Appendix C

Configuration Fields and Resolved Run Settings

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Appendix D

Model Hyperparameters and Training Commands

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Appendix E

Additional Figures and Tables

[STAGE 4 DRAFT REQUIRED]

[FIGURE REQUIRED]

[TABLE REQUIRED]

[VALIDATION REQUIRED]

Appendix F

Reproducibility and Environment Notes

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Appendix G

Legacy Code and Excluded Artifacts

[STAGE 4 DRAFT REQUIRED]

[VALIDATION REQUIRED]

Bibliography

- [1] Judea Pearl. “Causal Diagrams for Empirical Research.” In: *Biometrika* 82.4 (1995), pp. 669–688. DOI: 10.1093/biomet/82.4.669. URL: <https://doi.org/10.1093/biomet/82.4.669>.
- [2] Miguel A. Hernan and James M. Robins. “Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available.” In: *American Journal of Epidemiology* 183.8 (2016), pp. 758–764. DOI: 10.1093/aje/kwv254. URL: <https://doi.org/10.1093/aje/kwv254>.
- [3] Miguel A. Hernan and Sarah L. Taubman. “Does Obesity Shorten Life? The Importance of Well-Defined Interventions to Answer Causal Questions.” In: *International Journal of Obesity* 32.Suppl 3 (2008). Author/academic repost used because publisher PDF was not directly downloadable in this environment., S8–S14. DOI: 10.1038/ijo.2008.82. URL: <https://doi.org/10.1038/ijo.2008.82>.
- [4] Ikaro Silva et al. “Predicting In-Hospital Mortality of ICU Patients: The PhysioNet/Computing in Cardiology Challenge 2012.” In: *Computing in Cardiology*. Vol. 39. 2012, pp. 245–248. URL: <https://physionet.org/content/challenge-2012/1.0.0/>.
- [5] Alistair E. W. Johnson et al. “MIMIC-III, a freely accessible critical care database.” In: *Scientific Data* 3.160035 (2016). DOI: 10.1038/sdata.2016.35. URL: <https://doi.org/10.1038/sdata.2016.35>.
- [6] Hrayr Harutyunyan et al. “Multitask Learning and Benchmarking with Clinical Time Series Data.” In: *Scientific Data* 6.1 (2019), p. 96. DOI: 10.1038/s41597-019-0103-9. URL: <https://doi.org/10.1038/s41597-019-0103-9>.